



How Exoskeletons Help Reclaim Mobility

The introduction of exoskeletons has given hope to people with physical or neurological disorders by enabling them to walk, sit, or stand without any help. Exoskeletons are built to enhance the performance of humans working in fields where excessive human labour is needed. This article covers the role of exoskeletons in the healthcare industry and rising demand among people



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In the 2014 FIFA World Cup held at Brazil, Juliano Pinto, who is paralysed from the chest down, had performed his first kick in the opening ceremony, which showcased the advances in neuroengineering. The mastermind behind this mind-blowing scientific demonstration was the director and neuroscientist of Duke University Center for Neuroengineering, Dr Miguel Nicolelis, who is also leading the Walk Again project.

The goal of this project is to explore how brain-machine interfaces (BMIs) that incorporate virtual reality (VR) systems use patients' brain activity to control their legs and restore their mobility, strength, and independence.

With the help of BMI, Juliano was able to control the movements of the lower-limb robotic exoskeleton. Such ambulatory

exoskeletons are more effective in restoring and improving the motor functions of the user. Dr Nicolelis has eight patients under this Walk Again project. They are not only walking with the help of robotic exoskeleton legs but have now also learned to move forward by themselves.

Exoskeletons

Exoskeletons, as the name indicates, are skeletons existing on the outside of bodies with articulated joints similar to humans, and each joint is powered by one or more servomotors with sensor outputs used as rehabilitation or augmentation medical devices.

These exoskeletons are built to enhance the performance of humans working in fields like defence, logistics, or automotive industry, where excessive human labour is needed. Apart from improving the power and efficiency of individuals, these exoskeleton suits or power gloves support weight and reduce the person's effort, thereby reducing fatigue and stress. Hardiman, a hydraulic and electric heavy bodysuit for military use, was the first exoskeleton built by General Electric in the year 1965. The film industry has also used many exosuits that have made us think critically about their usage in practical life.

Exoskeletons in healthcare

The developments in artificial intelligence (AI), actuators, sensors, computer processors, 3D printers, and li-ion batteries have given a futuristic perspective in building exoskeletons similar to that used in science fiction. Also, newly developed power sources have increased the independence



Fig. 1: Award winning physicist Stephen Hawking strolling the Cambridge University in his exoskeleton suit